

component manufacturing, such as cables, displays, and PCBs. Together, they aim to build a complete electronics ecosystem. Innovation will be key not only in chip design but also in packaging, assembly, and system integration.

However, funding remains a challenge for Indian semiconductor startups. Last year, the government focused on manufacturing, including fabs and packaging. This year, the focus is shifting towards supporting startups through schemes promoted by MeitY. The DLI has increased funding and is also encouraging private investment.

The stated target is to fund 100 startups and scale at least five to \$100 million in revenue within five years. The long-term goal is for two Indian companies to enter the global top 10 in a decade. Achieving this will require a complete ecosystem extending beyond just design talent.

Many Indian startups are design-focused, but they struggle to convert their ideas into final products. They often become IP companies due to the lack of manufacturing support. In contrast, Taiwan has produced numerous chip companies with significant revenues, including MediaTek. Their success stems from experience across the full value chain, something India must now develop.

Crossing the 'valley of death' from design to full product development requires infrastructure, funding, IP support, and partnerships. Collaborating with experienced Taiwanese companies could help accelerate this journey.

Approach to manufacturing

India must take incremental steps, starting with OSAT (outsourced semiconductor assembly and test), and testing is a pragmatic approach. These require lower investment and are comparatively less complex. Full-scale fabs can follow. There is no

immediate need to pursue advanced nodes such as 3nm or 4nm. Mature nodes like 130nm and 180nm are more suitable starting points. These technologies are easier to implement and could succeed within a three- to five-year timeline.

The idea of microfabs, requiring investments of under \$10 million, across multiple locations could raise awareness and encourage hands-on experience. These fabs can produce non-logic components, such as sensors or power devices, which are making them ideal for learning and experimentation.

India's education system must also shift to enable more practical learning. Engineering students should be encouraged to tinker, build, test, and debug real circuits. This shift in mindset, alongside robust manufacturing, is essential.

Building for India and the world

Demand should not be a concern. While the market is global, success begins at home. Products must first solve local problems, be reliable, and easy to maintain. Once proven locally, they can be relevant globally. India has the opportunity to create solutions that are not only affordable but genuinely needed worldwide.

India stands at a semiconductor inflexion point, reminiscent of the 1980s when transistors became commercially viable. Now is the time to create millions of innovative products. The government is enabling this journey. For young engineers, this is a rare and exciting opportunity to contribute to a global transformation rooted in India. **EFY**

This article is based on a session titled 'Opportunities for Entrepreneurs in the Indian Semiconductor Industry,' delivered by Raja Manickam, Founder and CEO of iVP Semiconductor, at IEW 2025, held at the KTPO Expo Centre, Bengaluru. It has been transcribed and curated by Shubha Mitra, journalist at EFY.

Anyone who stops learning is old, whether at twenty or eighty.

—Henry Ford

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